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SUPERSEDING
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MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS, SWITCHING

TYPE 7849

This specification is approved for use by the Engineering and Technical Support Division (AFLC/LOIE), Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: TR, C-Band, 5395-5905 MHz.

ABSOLUTE RATINGS:

Parameter:	Incident Power pi	Ebb	Ii	T
Unit:	kw	Vdc	µAdc	°C
Maximum:	500	---	200	+60
Minimum:	---	-700	100	-29

PHYSICAL CHARACTERISTICS:

Dimensions:	See outline drawing
Weight:	1 1/2 pounds approximately

TEST CONDITIONS:

Parameter:	Incident Power pi	VSWR (max)	Ii	Ri	tp_1	tp_2	prr
Unit:	kw	----	µAdc	Meg	µs	µs	pps
Tolerance:	±10%	----	---	5%	±1.0%	0.05	±1.0%
1	30	1.10	100	4.0	1.0	0.5	1000
2	40	----	100	4.0	1.0	---	1000
3	4.0	----	100	4.0	1.0	---	1000
4	250	----	100	4.0	4.5	---	1000

Test Frequencies Tolerance ±.1 MHz
F_1 = 5395 MHz
F_2 = 5650 MHz
F_3 = 5905 MHz

GENERAL:

First Article Inspection - Required (See Note 5)

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, Part 1						
1027	Temperature cycling nonoperating	-		---	---	---	---
4401	Ignitor ignition time	-	Ebb = -700 Vdc Ri = 4.0 Meg	t	---	5.0	sec
4406	Ignitor voltage drop	-	Ii = 100 µAdc See Note 1	Eid	-200	-400	Vdc
4416	Insertion loss	-	F = F_2 ±0.1% See Note 6	Li	---	0.7	db
4471	Recovery time	4	F = F_2	t	---	8	µs
4452	Flat leakage power (1)	1	F = F_2	pf	---	50	mw
4452	Spike leakage energy (1)	1	F = F_2	Ws	---	0.25	erg
4473	Voltage standing wave ratio	-	F_1 ± 0.1%	σ	---	1.9	---
		-	F_2 ±0.1%	σ	---	1.4	---
			F_3 ±0.1% See Note 6	σ	---	1.9	---
4488	Arc loss (1)	2	F = F_2	La	---	0.8	db
	Quality conformance inspection, part 2						
4488	Arc loss (2)	3	F = F_1 and F_3	La	---	0.8	db
4452	Flat leakage power (2)	1	F = F_1 and F_3	pf	---	50	mw
4452	Spike leakage energy (2)	1	F = F_1 and F_3	Ws	---	0.25	erg
4496	Firing power	3	F = F_2	Po/Du	---	150	mw
1031	High-frequency vibration	-	t = 5 mins See Note 2	---	---	---	---
4474	High-level VSWR	2	F = F_2	VSWR	---	1.10	---
----	Noise figure change	-	F = F_2 Ebb = -700 Vdc Ii = 100 µAdc T = 25°C pi = 1.0 mw See Note 3	ΔNF	---	0.7	db
----	TR noise ratio	-	F_1, F_2, and F_3 Ii = 100 µAdc See Note 4	ΔNF	---	1.3	---

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3						
----	Life test	4	Group D F = F_2	t	---	1000	hrs
	Life test end points:						
4471	Recovery time	4	F = F_2	t	---	15.0	μs
4452	Flat leakage power	1	F = F_2	pf	---	70	mw
4452	Spike leakage energy	1	F = F_2	Ws	---	0.35	erg
4416	Insertion loss		Ii = 100 μAdc	Li	---	1.0	db

NOTES:

1. The recommended ignitor operating current is for a tube with an average ignitor voltage drop. The following formula should be used to determine the value of the required series resistance for test.

$$Ri = \frac{Ebb - Eid}{150\ \mu Adc}$$

where Ri = Total Series resistance (megohms)
Ebb = Open circuit supply voltage (volts)
Eid = Average (center) ignitor voltage drop (volts)

At least 0.5 megohm of the total should be located as close as possible to the ignitor top cap to prevent oscillation.

2. Vibrate in a plane normal to the ignitor electrode. There shall be no evidence of shorts between keep-alive electrode and body as indicated by a short indicator during vibration. The tube shall satisfy all applicable requirements of this specification after vibration.

3. The noise figure change shall be the increase in the noise figure which occurs when the tube is placed between the noise source and the crystal detector compared to the noise figure before insertion of the tube.

4. The tube shall have an equivalent noise ratio not exceeding the specified value, as determined from the following equation:

$$t_{TR} = 1 + \frac{\text{antilog } (0.1F)\ \text{Antilog } (0.1Y-1)}{a}$$

where F = Receiver noise figure in db

Y = IF attentuation required to reduce the output noise level with ignitor ON to that recorded with the ignitor OFF.

a = 1 for a circuit with image rejection, or
2 for a circuit without image rejection.

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NOTES: -Continued

5. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this specification sheet. In addition, first article sample approval shall include satisfactory demonstration of tube-to-system compatibility. Invitation for bids should provide that the preparing activity reserves right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6. A swept frequency may be used for this test.

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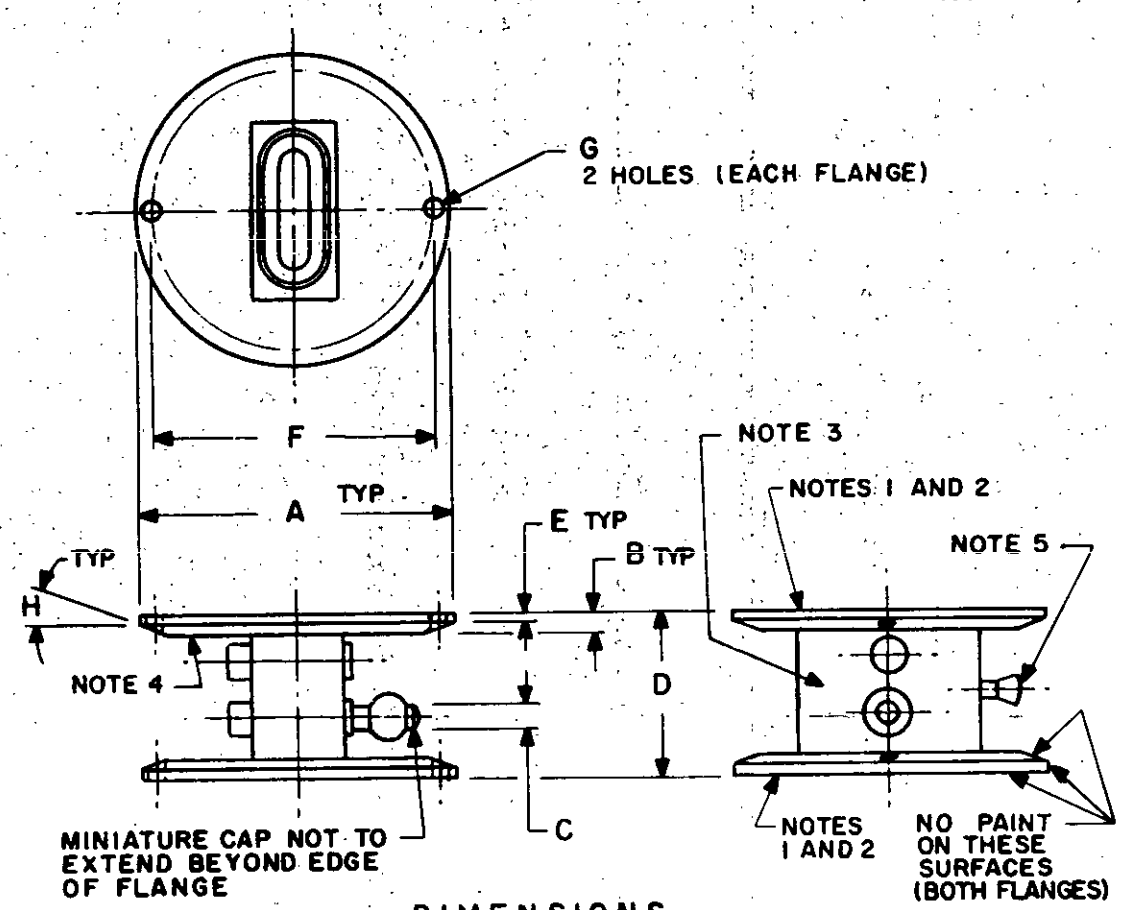
Preparing activity:
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User activity:
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DIMENSIONS

SYM	INCHES		MILLIMETERS		SYM	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
A	3.610	3.640	91.69	92.46	E	.100	.120	2.54	3.05
B	.235	.265	5.97	6.73	F	3.245	3.255	82.42	82.68
C	.245	.255	6.22	6.48	G	.213	.223	5.41	5.66
D	1.935	1.955	49.15	49.66	H	20°		20°	

FIGURE 1 Outline Drawing for Tube Type 7849

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NOTES:

1. Silver Plate Flange Faces per MIL-F-14072.
2. Rhodium Plate .000005 to .00001 thick on Flange Faces.
3. Paint Finish to be gray, per MIL-E-15090B, Type 2, Class 2 Air Dry, Color 26329 of Federal Standard 595, on all surfaces except flange faces, as noted.
4. Part to be marked "High Power Side" in 7/64" high character color white, on the inside of the Input Flange.
5. Exhaust cap not to extend beyond flanges.
6. Nickel plate may be used in lieu of plating specified in note 1 and note 2.

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